the small: the realm of atoms, molecules, electrons, and protons. In this realm, the Newtonian laws had been found to fail in ways that relativity had not anticipated. Their replacement was a radically new set of physical laws —laws that were called *quantum mechanics*[1] because they deal with the behavior (the "mechanics") of particles of matter ("quanta"). Though only two years old, the new quantum mechanical laws were already having great success in explaining how atoms and molecules behave.

Chandrasekhar had read in Sommerfeld's book about the first, tentative version of the new laws. But the tentative quantum laws had been unsatisfactory, Sommerfeld explained to him. Although they agreed well with experiments on simple atoms and molecules such as hydrogen, the tentative laws could not account for the behaviors of more complicated atoms and molecules, and they did not mesh in a logically consistent way with each other or with the other laws of physics. They were little more than a mishmash of unaesthetic, ad-hoc rules of computation.

The new version of the laws, though radical in form, looked far more promising. It explained complicated atoms and complicated molecules, and it seemed to be meshing quite nicely with the rest of physics.

Chandrasekhar listened to the details, entranced.

Quantum Mechanics and the Guts of White Dwarfs

When they parted, Sommerfeld gave Chandrasekhar the galley proofs of a technical article that he had just written. It contained a derivation of the quantum mechanical laws that govern large collections of electrons squeezed together into small volumes, in a metal for instance.

Chandrasekhar read Sommerfeld's galley proofs with fascination, understood them, and then spent many days in the University library studying all the research articles he could find relating to them. Especially interesting was an article entitled "On Dense Matter" by the English physicist R. H. Fowler, published in the 10 December 1926 issue of *Monthly Notices of the Royal Astronomical Society.* Fowler's article pointed Chandrasekhar to a most fascinating book, *The Internal Constitution of the Stars,* by the eminent British astrophysicist Arthur S. Eddington, in which Chandrasekhar found a description of the *mystery of white-dwarf stars.*